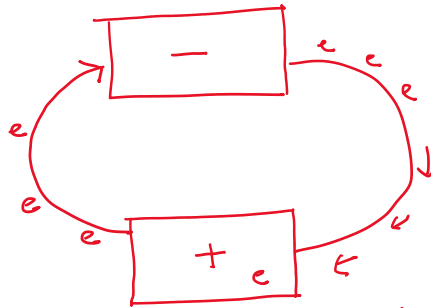




Separation of charges

+ -

↘ electromotive force

1. EMF: Separation of charges (Voltage diff)
2. Flow of charge: current

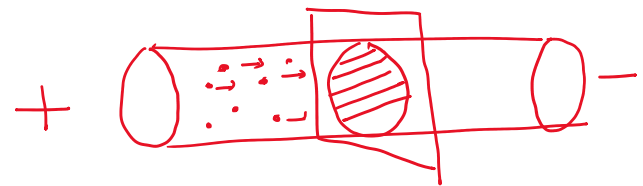
$$I = \frac{dq}{dt}$$

$$i(t) = \frac{dq(t)}{dt}$$

$$= dq(t) = i(t)dt \Rightarrow dQ = i dt$$

$$\int dQ = \int_t i dt$$

$$\therefore Q(t) = \int_{t_0}^t i(t) dt$$

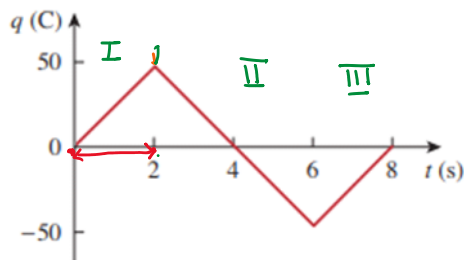


DC



$$I' = -I = -5A$$

$$i(t) = \frac{dq(t)}{dt} \quad \begin{matrix} y \rightarrow q \\ x \rightarrow t \end{matrix}$$



$$I: \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\Rightarrow y - 0 = \frac{0 - 50}{0 - 2} (x - 0)$$

$$\Rightarrow y = 25x$$

ID.

$$\Rightarrow q = 25t \quad \text{For } 0 < t \leq 2s$$

$$i(t) = \frac{dq(t)}{dt} = \frac{d}{dt}[25t] = 25A \quad 0 < t \leq 2s$$

Region II: $2 < t < 4$

$$y - y_1 = \frac{y_1 - y_2}{x_1 - x_2} (x - x_1)$$

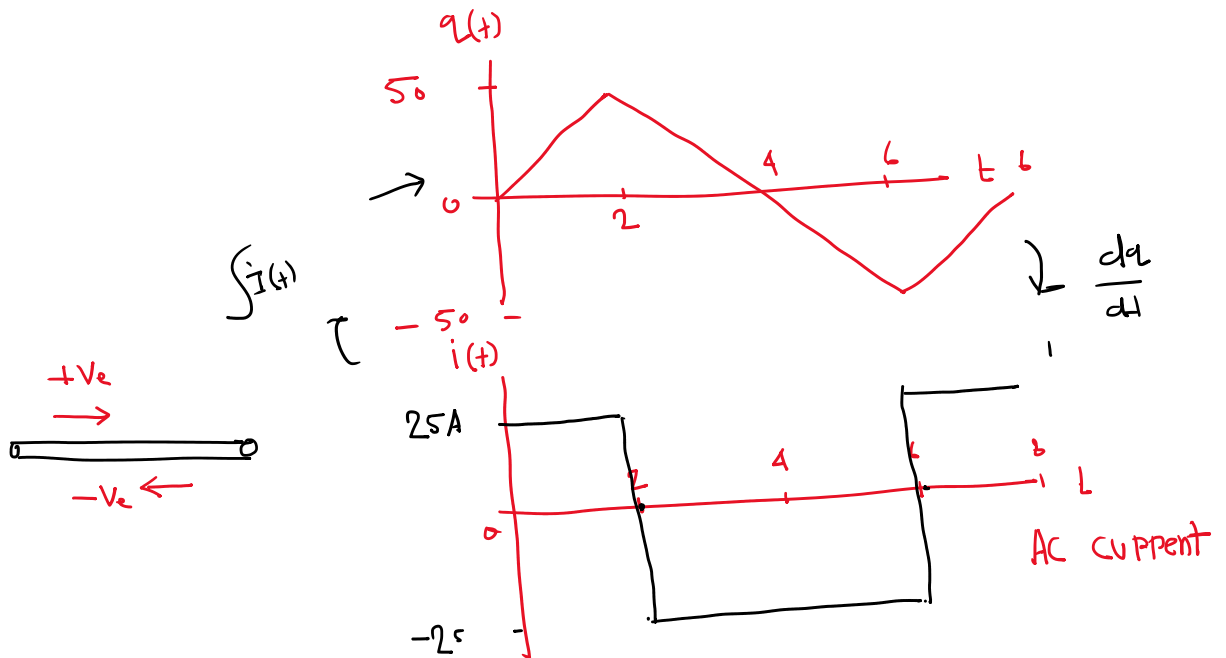
$$\Rightarrow y - 50 = \frac{50 - 0}{2 - 4} (x - 2) \Rightarrow y - 50 = -25(x - 2)$$

$$\Rightarrow y = -25x + 50 + 50$$

$$\Rightarrow y = -25x + 100$$

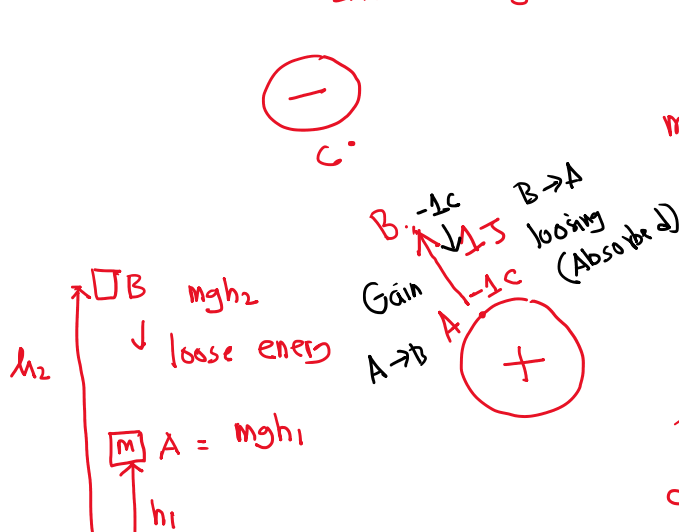
$$\Rightarrow q = -25t + 100 \quad 2 < t < 6$$

$$i(t) = \frac{dq}{dt} = \frac{d}{dt} [-25t + 100] = -25 \text{ A}$$



$$\frac{1 \text{ C}}{1 \text{ s}} = 1 \text{ A} \quad \text{C/s}$$

EMF: Voltage difference



@ position A: -1 C charge
move -1 C charge from A to B.

Apply force $\rightarrow$ work must be done on the charge $\rightarrow$ Energy must be expended to move -1 C charge from A $\rightarrow$ B.

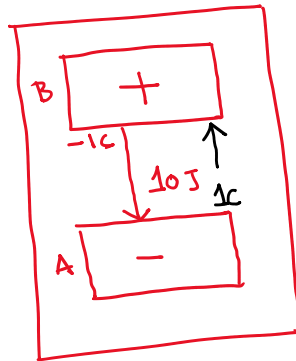
If the amount of the work is 1 J then we say that there is voltage difference betn B & A is 1 V .



difference betn B & A is 1V.
 $V_{BA} = 1V$ or the potential difference betn B & A is $V_{BA} = 1V$

$$V_{CA} = 2V$$

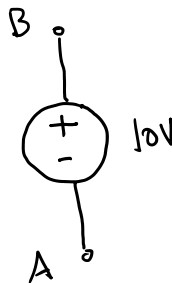
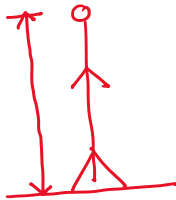
$$V = \frac{W}{Q} \quad \text{Amount of work done / C charge}$$



$$V_{BA} = 10V$$

-1C charge from B to A it requires 10J.

Voltage/Potential difference always measured between two Points.



$$V_{AB} = -10V$$



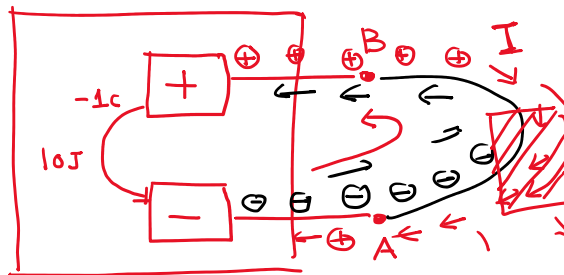
Ref: A →

Ref can be assigned to 0V

$$V_{BA} = V_B - V_A = V_B - 0$$

$$V_{BA} = 10V$$

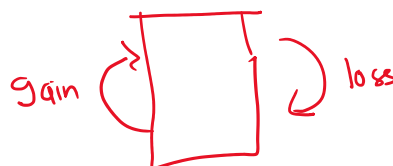
Cause Energy gain



Flow of +ve charge is current

Response loss of energy

Bulb → glows



$$\frac{W}{Q} = V \quad \frac{Q}{t} = I$$

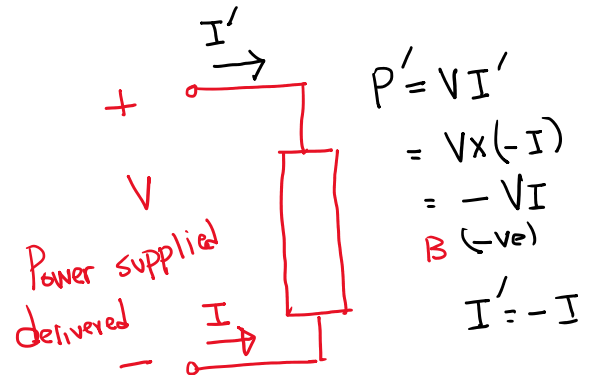
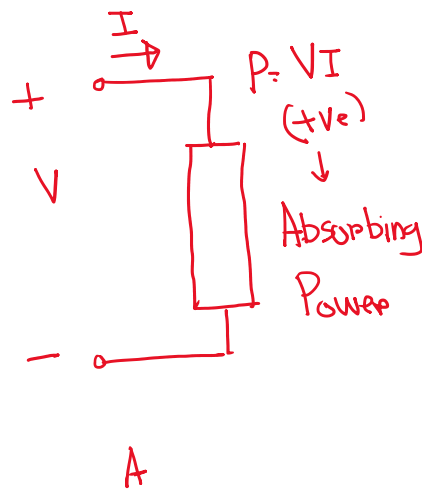
to move 1C charge in 1s : 1Watt

1 Joule expended/used to move 1C charge in 1s : 1 Watt

$$P = \frac{dW}{dt} = \frac{dW}{dq} \frac{dq}{dt} = VI$$

$P = VI$: Voltage time current

- Two types of element:
- Active element: Supplies energy
Voltage source, current source
 - Passive element: Absorbs energy
Resistor, capacitor, inductor, voltage source



If current is entering the -ve terminal, then assign a current as if its entering the +ve terminal of the voltage. then determine power